GAME THEORY & NASH EQUILIBRIA — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Game Theory & Nash Equilibria

Question 1

10.0 points

Examine the extensive form game tree in the diagram below for a sequential two-player game where Player 1 can choose between actions A and B, and Player 2 can then choose between X and Y after observing Player 1's action. Identify the subgame perfect Nash equilibrium by performing backward induction on this game tree.

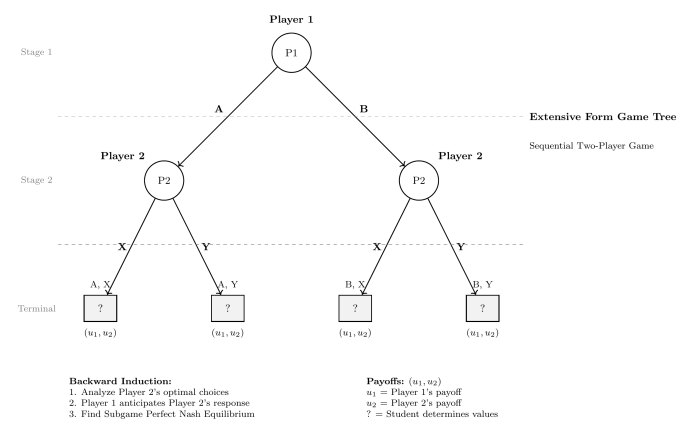

Fig. 1. Extensive form game tree for a sequential two-player game. Player 1 chooses between actions A and B. Player 2 chooses between actions X and Y after observing Player 1's action. Terminal nodes show payoffs for each action combination. IMPORTANT: The diagram is missing required labels. The diagram is missing required labels for ". Add each missing label as visible text at the appropriate location in the diagram.

Question 2

10.0 points

Consider the payoff matrix for a Prisoner's Dilemma shown in the diagram below: | | Cooperate | Defect | |-------|----------|--------| | C | (3,3) | (0,5) | | D | (5,0) | (1,1) | Identify all Nash equilibria in this game and explain why they are Nash equilibria.

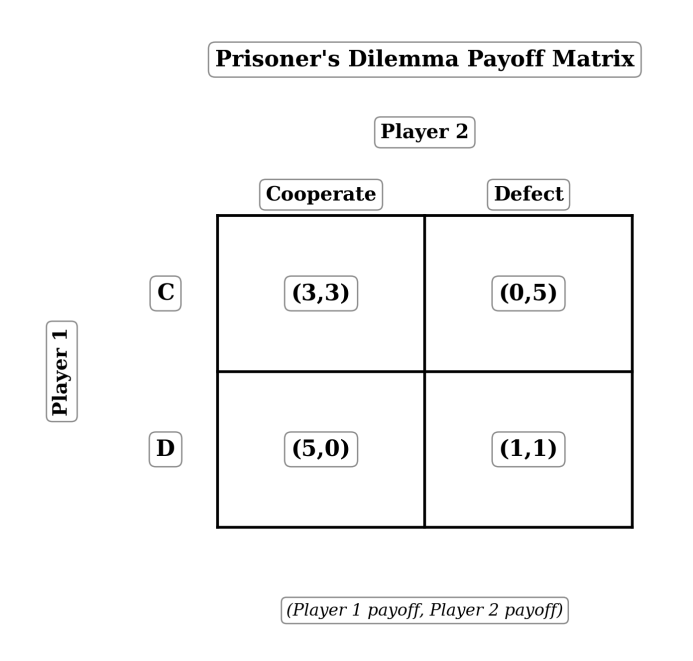

Fig. 2. A payoff matrix for a Prisoner's Dilemma game with two players. The matrix has two strategies for each player: Cooperate and Defect. The payoffs are as follows: (3,3) for both cooperating, (0,5) if player 1 cooperates and player 2 defects, (5,0) if player 1 defects and player 2 cooperates, and (1,1) if both defect. The matrix should be clearly labeled with strategies and payoffs. IMPORTANT: The diagram is missing required labels. The diagram uses abbreviations 'C' and. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram is missing the full text labels for. Add each missing label as visible text at the appropriate location in the diagram.

Question 3 10.0 points

Evaluate a mixed strategy Nash equilibrium in a coordination game given by the payoff table in the diagram below, where both players choose high or low effort: | | High | Low | |----------|-------|-------| | High | (2,2) | (0,3) | | Low | (3,0) | (1,1) | Use the mixed strategy method to find the equilibria, and determine the probabilities with which each player should randomize their strategies.

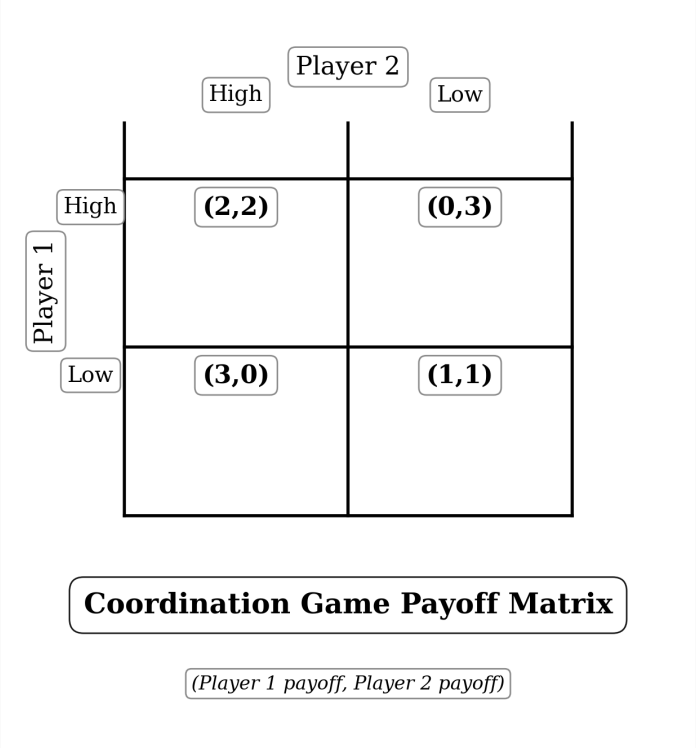

Fig. 3. Payoff matrix for a coordination game with two players. Player 1's strategies are rows (High, Low) and Player 2's strategies are columns (High, Low). Payoffs are (2,2) for both choosing High, (0,3) for Player 1 choosing High and Player 2 choosing Low, (3,0) for Player 1 choosing Low and Player 2 choosing High, and (1,1) for both choosing Low. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows a payoff of (3,0. Regenerate using EXACTLY the values and sequences specified.

Question 4

10.0 points

Using the diagram below, perform backward induction on the game tree for a three-stage sequential game with perfect information. Each player has one possible optimal move at each decision node. Determine the optimal strategy for each player, and explain how backward induction aids in determining this strategy.

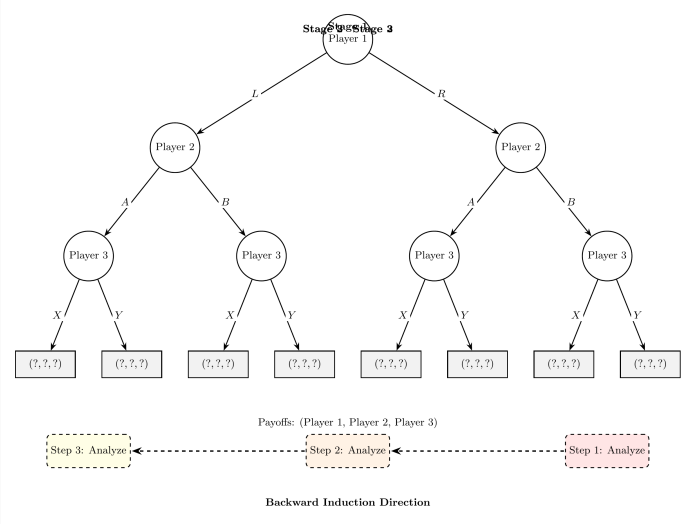

Fig. 4. A three-stage sequential game tree with perfect information, showing decision nodes for each player and branches representing possible moves. Each player has one optimal move at each decision node. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram is unsolvable because it omits the. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram is unsolvable because it omits. Regenerate using EXACTLY the values and sequences specified.

Question 5 10.0 points

Examine the Pareto frontier in the diagram below, which illustrates efficient versus inefficient outcomes. Identify the Pareto efficient points and describe the trade-offs at inefficient points. Explain how any shift in the curve can alter efficiency in allocation.

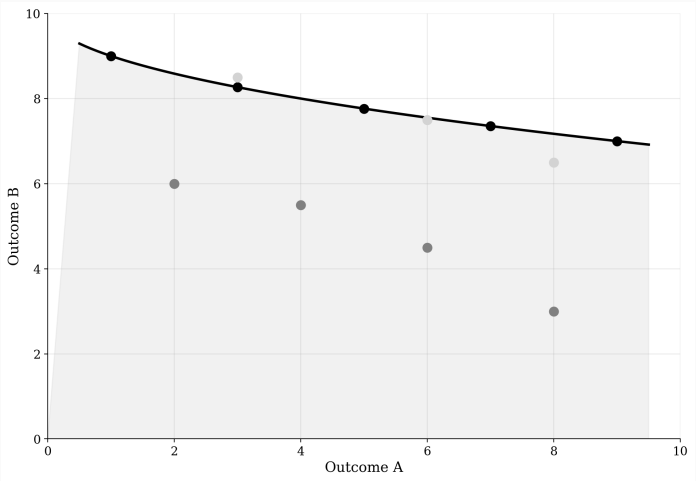

Fig. 5. A diagram illustrating a Pareto frontier with efficient and inefficient points. The curve represents the frontier, with points on the curve labeled as Pareto efficient and points below the curve labeled as inefficient. Include arrows to indicate potential shifts in the curve and how they might alter efficiency in allocation. Label axes appropriately to show the trade-offs between two economic outcomes, such as 'Good A' and 'Good B'. Highlight the efficient frontier and inefficient regions clearly. IMPORTANT: The diagram is missing required labels. The diagram description requires arrows to indicate potential shifts in. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram is missing several required labels specified in the. Add each missing label as visible text at the appropriate location in the diagram.